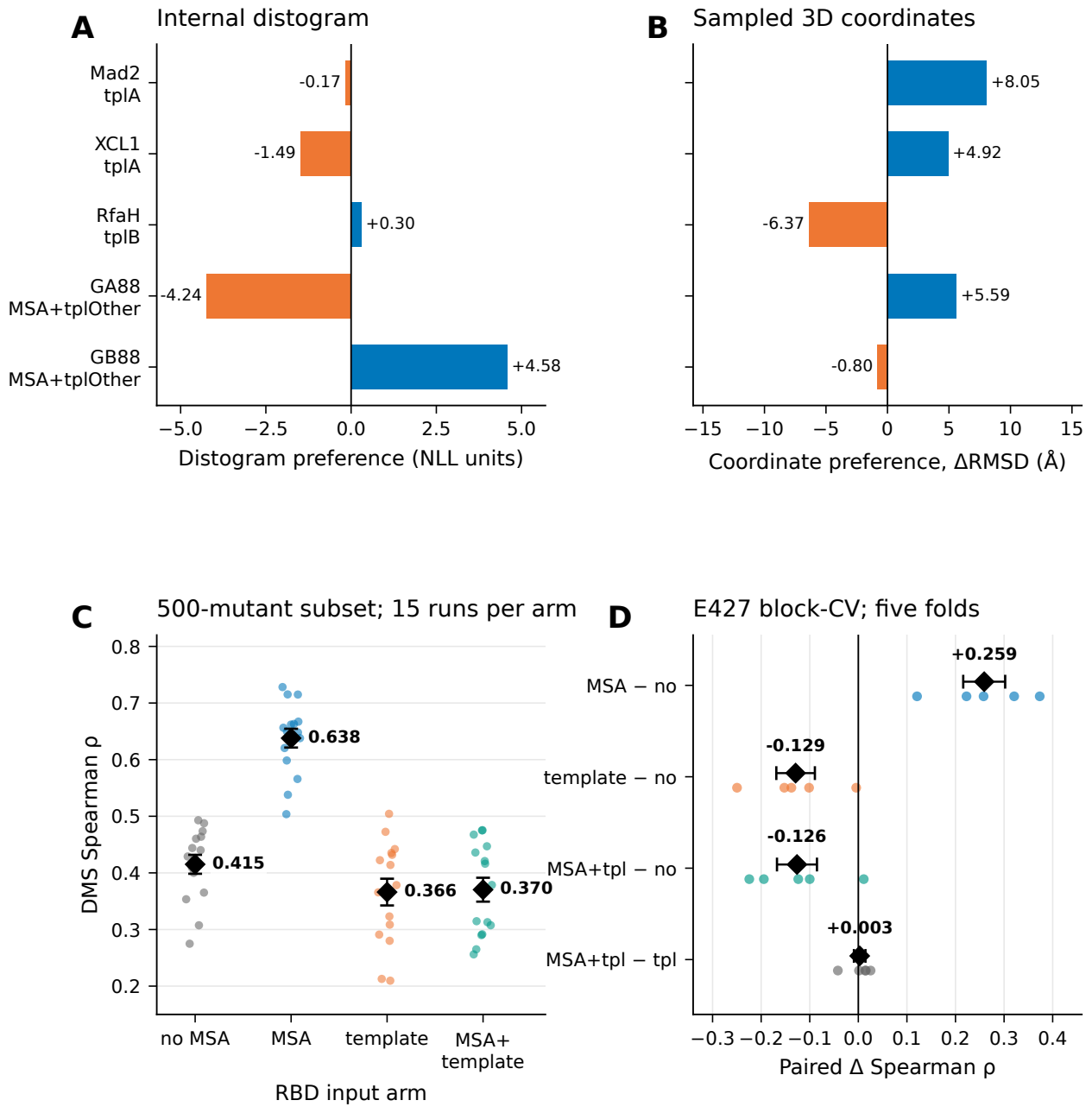

A
Internal distogram
Mad2 tplA
XCL1 tplA
RfaH tplB
GA88 MSA+tplOther
GB88 MSA+tplOther
-0.17
-1.49
+0.30
-4.24
+4.58
−5.0
−2.5
0.0
2.5
5.0
Distogram preference (NLL units)
B
Sampled 3D coordinates
+8.05
+4.92
-6.37
+5.59
-0.80
−15
−10
−5
0
5
10
15
Coordinate preference, ΔRMSD (Å)
C
500-mutant subset; 15 runs per arm
DMS Spearman ρ
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.415
0.638
0.366
0.370
no MSA
MSA
template
MSA+ template
RBD input arm
D
E427 block-CV; five folds
MSA − no
template − no
MSA+tpl − no
MSA+tpl − tpl
+0.259
-0.129
-0.126
+0.003
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Paired Δ Spearman ρ